Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/21/2025
PI Name:	Norcross, Jason M.S.		
Project Title:	Validation of Fitness for Duty Standards Using Pre- and Post-Flight Capsule Egress and Suited Functional Performance Tasks in Simulated Reduced Gravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-7114
Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	12/01/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/1/2027 per HHC element/JSC (Ed., 12/14/20)		
Key Personnel Changes/Previous PI:	There have been no significant changes to the study team over the past year.		
COI Name (Institution):	Marshall-Goebel, Karina Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Schlotman, Taylor (KBR) Peters, Brian (KBR) Cox, Lauren (JES Tech) Strock, Nicole Ph.D. (KBR)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Rigorous adherence to available inflight countermeasures has effectively mitigated losses or maintained muscle strength and aerobic capacity in some returning long-duration International Space Station (ISS) crewmembers; however, all astronauts demonstrate significant decrements in functional performance upon return to a gravity environment. These losses in functional performance can be largely attributed to neurovestibular / sensorimotor deficits that can take days or weeks from which to recover and for which there is no current operational countermeasure. Although these losses are tolerable for current land-based returns to Earth, where ground personnel can quickly support the crew at the landing site, this will not be the case for future off-nominal water-based Orion landings or for nominal Mars surface landings, both of which will require crewmembers to be capable of egressing their landing vehicle unassisted. Quantification of astronauts' post-landing functional capacity including ability to perform an unassisted capsule egress and critical planetary extravehicular activity (EVA) tasks is necessary to design concepts of operation for Moon and Mars exploration mission systems and ultimately to promote exploration mission success. These results can then be reviewed in combination with other pre-flight, in-flight, and post-landing measures and determinants of health and performance (e.g., sleep, nutrition, exercise) to help develop and select necessary countermeasures capable of protecting all crewmembers or to identify characteristics (both behavioral and inherent) that might allow for selection of crew dependent on mission objectives. Data collected in this proposal will provide unique data on unassisted capsule egress while either shirt-sleeved or wearing an unpressurized launch, entry, abort (LEA) suit in Earth's gravity and on EVA-relevant functional task performance by testing astronauts shortly after return to Earth while suited and pressurized in a simulated reduced gravity analog. The research product will be a temporal profile of unassisted capsule egress and planetary EVA task performance pre-flight and at multiple post-landing intervals, the timing of which will be determined based on post-landing logistics and coordination with other investigations. Data will be collected for spaceflight missions ranging from 2 months, 6 months, and up to 1 year in duration. Results of the proposed study will be used in combination with subsequent definition and design of exploration mission systems and operations concepts to define data-based Fitness for Duty standards.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The core focus of this study is to facilitate safe exploration of Mars and return back to Earth, which albeit has limited direct impact to the people on Earth, but to the extent that space exploration is for the good of all humanity, this study will facilitate more successful missions to Mars.
Task Progress:	The Institutional Review Board (IRB) Approval has been completed and maintained for the pilot study (JSC eIRB STUDY00000242) and CIPHER Egress Fitness (JSC eIRB STUDY00000169) and test readiness review (TRR) documentation has been completed and maintained for both studies. Pilot Egress Fitness was amended to include the possibility of collecting post-flight capsule egress data with commercial astronauts. To date, Pilot Egress Fitness has enrolled 12 total subjects (4 complete, 2 dropped, 2 currently inflight, 4 consented and awaiting initial data collection). CIPHER Egress Fitness has enrolled 8 total subjects, with 1 withdrawal and 3 subjects now complete, but with only 1 subject able to complete every test session due to various facility and schedule reasons. Three subjects have initiated pre-flight baseline data collection and 1 additional subject has consented but has not started. The update to the capsule egress mockup was discussed in the last Task Book update, but it has now been used for actual testing and a second copy of the capsule egress mockup has been built to provide more flexibility in covering different landing sites. Beyond the updates to the capsule mockup, there are no new data or protocol changes to report on since the last Task Book report in February 2024. Schedule Update: CIPHER Egress Fitness has enrolled 8 total subjects, with 3 complete and 1 withdrawal. ARGOS was returned to operations in March 2024 allowing for necessary postflight testing of the 2 subjects and preflight testing for 3 others. One subject was limited to capsule egress only. Pilot Egress Fitness has re-opened to enrollment and waiting to complete data collection on 6 subjects spread over 2 different spaceflights.
Bibliography Type:	Description: (Last Updated: 01/08/2026)
Abstracts for Journals and Proceedings	Norcross JR, Schlotman TE, Cox LI, Rhodes R, Strock N, Young M, Marshall-Goebel KJ. "Validation of fitness for duty standards using pre- and postflight capsule egress and suited functional tasks in simulated reduced gravity." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024